

CLOUDCORE SOLUTIONS

Advanced Cloud & Data Analytics

RESPONSE TO REQUEST FOR PROPOSAL

Migration and Modernisation of Manga Data Architecture to the Cloud

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Cloud Analytics — Group Work Option B

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1. EXECUTIVE SUMMARY

Manga stands at an inflection point. With **annual revenues of €280 million** and a physical presence across **100+ stores worldwide**, the brand has built a strong foundation. Yet its growth rate of **only 3.5%** falls short of the market average, online sales grew a mere **1.2% year-over-year**, and the company currently spends **twice as much as competitors** while achieving half the analytical output. The root cause is architectural: independent ETL pipelines triggered by daily cron jobs, aggregating into Excel files shared once per day, cannot power the real-time, personalised, data-driven experiences that modern retail demands.

CloudCore Solutions proposes the **Manga DataHub** — a cloud-native AWS data lakehouse that replaces this legacy infrastructure with a unified, scalable, and secure platform built for the next decade of retail growth. Our proposal is structured around three strategic pillars:

- **Unify** — Centralise all data sources into a single governed platform accessible across every department, replacing siloed pipelines and manual Excel reporting.
- **Accelerate** — Reduce time-to-insight from 24 hours to sub-second, enabling real-time inventory management, live sales dashboards, and instant fraud detection.
- **Enable** — Deliver the ML-powered features on Manga's roadmap — product recommendations, dynamic pricing, demand forecasting, and the virtual shopping assistant — that are simply impossible with the current architecture.

The proposed solution is built entirely on AWS using open-source tooling (Apache Airflow, Apache Spark, open Parquet formats) to ensure portability and eliminate vendor lock-in. Three fully isolated environments (Development, Pre-Production, Production) are provisioned from identical Infrastructure-as-Code templates, with anonymised production data samples used across non-production environments. Security is designed in from day one: column-level access controls, automated PII scanning, KMS encryption, and full GDPR audit trails address Manga's compliance requirements and reinforce the trust of its customers.

Expected outcomes within 12 months: **40–60% reduction in infrastructure costs** vs. the current colocation model; **real-time reporting replacing the daily batch cycle**; and the first ML-powered use cases — demand forecasting and product recommendations — **live within six months of project kickoff**.

Dimension	Current State	With Manga DataHub
Reporting lag	24 hours (daily batch)	Sub-second (real-time streaming)
Infrastructure cost	2× competitor average	Projected 40–60% reduction
BI access	Manual Excel files	Self-service dashboards
ML capabilities	None	Forecasting, recommendations, dynamic pricing
Data environments	Single (production only)	Dev / Pre-Prod / Prod (isolated)
GDPR compliance	Unclear	Full — automated PII scanning & masking

2. HIGH-LEVEL ARCHITECTURE

The Manga DataHub architecture is designed around five core layers, three cross-cutting concerns, and a clear separation between batch and streaming data paths. This section presents the conceptual architecture independent of specific technology choices, grounding each decision in foundational cloud analytics principles.



Figure 1 — Manga DataHub High-Level Architecture (Cloud-Agnostic)

Interactive version available at: <https://manga-cloud-platform.streamlit.app/> — click any component for design rationale and RFP requirement coverage.

2.1 Architectural Layers

Layer 1: Data Sources

Manga's data originates from four distinct system categories: Point-of-Sale and ERP systems generating structured transactional and inventory data; web and mobile applications producing continuous, semi-structured event streams (clickstream, reviews, session data); CRM and loyalty platforms holding customer demographic records; and external APIs providing contextual data including weather conditions, public holidays, logistics partner feeds, and marketing campaign results. The diversity of these sources — structured, semi-structured, and unstructured, at varying velocities — demands a platform capable of handling all modalities within a single unified architecture.

Layer 2: Ingestion

The ingestion layer replaces all existing cron-job ETL pipelines with two parallel paths. A **streaming ingestion path** handles real-time event feeds from POS transactions and web interactions, processing data within milliseconds of generation. A **batch ingestion path** handles scheduled, high-volume loads from CRM systems, inventory reconciliations, and external API pulls.

A third decoupled **API connector layer** enables integration with new external data sources without modifying existing pipelines, directly enabling extensibility (R7).

Layer 3: Storage — The Lakehouse

Data is stored in a three-zone lakehouse model. The **Raw Zone** holds immutable source data exactly as received — the system of record enabling full historical reprocessing. The **Curated Zone** contains cleaned, validated, deduplicated data with PII masked for non-production access. The **Serving Zone** provides pre-aggregated, query-optimised datasets for BI tools and ML model serving. This medallion architecture ensures clear data contracts between layers and enables different consumer types to access data at the appropriate level of transformation.

Layer 4: Processing

The processing layer performs transformation, orchestration, and governance across all data. A distributed computation framework applies business logic at scale. A DAG-based orchestration engine replaces all cron jobs with monitored, dependency-aware pipelines with built-in retry logic and alerting. Automated data quality checks run after every pipeline execution. A centralised data catalog maintains schema lineage, ownership, and discovery metadata for self-service access.

Layer 5: Consumption

Data surfaces to three consumer types. **BI and reporting** users access self-service dashboards replacing manual Excel reports. **ML and AI teams** access a managed model training, deployment, and monitoring environment with pre-built feature stores. **External consumers** — logistics partners, marketing platforms, and AI agents — access curated datasets via secure, versioned APIs.

2.2 Cross-Cutting Concerns

Three concerns apply across all layers and are not optional add-ons:

- **Security & Compliance:** Encryption at rest and in transit, RBAC at column and row level, automated PII detection and masking, full GDPR audit trails, and threat detection are built into every layer from inception.
- **Infrastructure as Code & Automation:** All infrastructure is version-controlled and provisioned programmatically. Three identical environment configurations (Dev, Pre-Prod, Prod) are deployed from a single set of templates.
- **Cost & Sustainability:** Auto-scaling matches compute resources to actual demand. Data lifecycle policies archive cold data automatically. Carbon emissions are tracked per service and region, providing quantifiable improvements versus the on-premises colocation model.

2.3 Design Principles

Principle	Application in Manga DataHub
Separation of Concerns	Each layer has a single responsibility. Storage is decoupled from compute.
Modularity	New data sources, jobs, and models are added as isolated modules. Failure in one does not cascade.
Elasticity	All compute scales automatically with workload. A Fashion Week spike requires no manual intervention.
Scalability	The architecture handles current volumes and the 10× growth from Manga's international expansion.
Open Standards	Parquet, Apache Airflow, Apache Spark — open formats and open-source tools at every layer. No lock-in.

3. LOW-LEVEL ARCHITECTURE

This section maps the conceptual architecture to specific AWS services, providing the rationale for each selection based on performance benchmarks, operational reliability, cost model, and alignment with Manga's requirements.

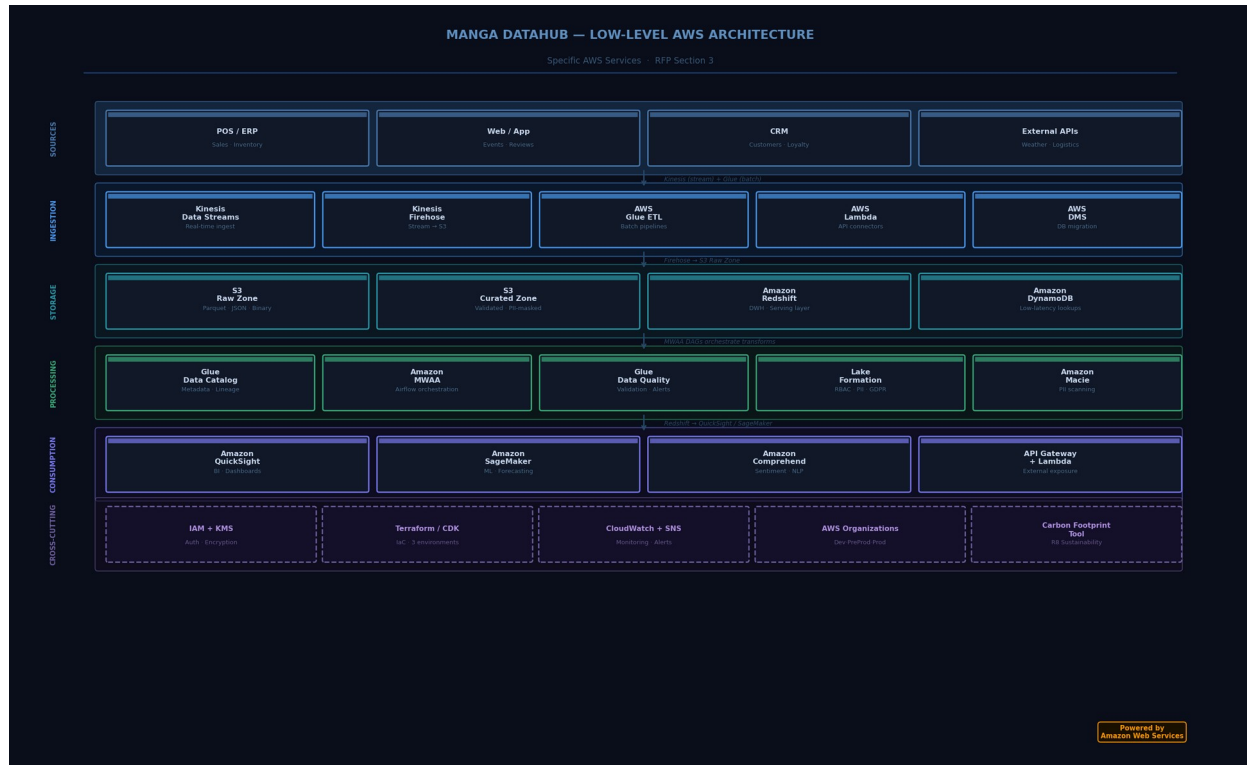


Figure 2 — Manga DataHub Low-Level AWS Architecture

Interactive service explorer: <https://manga-cloud-platform.streamlit.app/> — click any AWS service for pricing model, rationale, and the specific R1–R8 requirement it addresses.

3.1 Ingestion Layer

Amazon Kinesis Data Streams + Kinesis Data Firehose

Kinesis Data Streams provides managed, sub-second streaming ingestion for POS transaction events and web/app event data. It scales automatically across shards to handle variable throughput, including peak periods such as Fashion Week campaigns identified in Manga's external factors data. Kinesis Firehose buffers stream output and delivers micro-batches to S3 in Apache Parquet format, handling format conversion without additional code. **Pricing:** pay per shard-hour and per GB ingested. **Addresses:** R1 (streaming), R3 (automation).

AWS Glue ETL

Managed Apache Spark replaces all hand-coded ETL scripts for batch data paths including inventory reconciliation, CRM syncs, and historical sales loads. Built-in job bookmarks prevent duplicate processing on reruns. **Pricing:** pay per DPU-hour. Spot DPUs reduce batch job costs by up to 70%. **Addresses:** R1 (batch), R3 (automation).

AWS Lambda + Amazon EventBridge

Serverless functions handle all third-party API connector logic: polling the weather API, public holiday calendar, and logistics partner endpoints on schedule. New external data sources are

added as new Lambda functions without modifying existing pipelines. **Pricing:** pay per invocation. **Addresses:** R7 (extensibility).

AWS Database Migration Service (DMS)

DMS handles the one-time migration of historical data from the on-premises colocation databases to S3. Ongoing Change Data Capture (CDC) replication runs during the transition period to ensure zero data loss at cutover. **Addresses:** R3 (automation), R4 (high availability during migration).

3.2 Storage Layer

Amazon S3 — Three-Zone Data Lakehouse

S3 stores all data across the three-zone medallion architecture. **Raw Zone:** Immutable, append-only. Parquet for structured, JSON for events, binary for review images. S3 Intelligent-Tiering automatically moves objects to cheaper storage classes after 30 days of inactivity. **Curated Zone:** Schema-validated Parquet, partitioned by date, PII masked. **Serving Zone:** Pre-aggregated datasets optimised for analytical queries. Open format: zero lock-in. **Pricing:** as low as €0.004/GB/month with Glacier tiering. **Addresses:** R1, R6, R8.

Amazon Redshift

Columnar data warehouse serving as the Gold layer for BI and ad-hoc queries. Redshift Spectrum enables querying S3 data directly without loading, reducing storage duplication. RA3 nodes decouple compute and storage scaling. Replaces the daily Excel aggregation entirely. **Pricing:** Reserved Instances offer up to 75% discount vs on-demand. **Addresses:** R1, R6.

Amazon DynamoDB

NoSQL key-value store for sub-10ms lookups required by ML inference endpoints and the planned virtual shopping assistant. Stores product recommendation results and real-time inventory availability flags. **Pricing:** pay-per-request for variable workloads. **Addresses:** R1 (low-latency), R7 (API exposure).

3.3 Processing & Governance Layer

Service	Role	Open Source?	RFP Req.
AWS Glue Data Catalog	Central metadata, schema registry, lineage tracking, self-service discovery	Apache Hive Metastore	R1, R7
Amazon MWAA (Airflow)	DAG-based pipeline orchestration replacing all cron jobs; retry logic, SLA alerts, visual UI	Apache Airflow	R3
AWS Glue Data Quality	Automated quality rules (null checks, range validation, referential integrity, freshness); alerts	—	R5
AWS Lake Formation	Fine-grained RBAC at column and row level; centralised GDPR policy management; data sharing controls	—	R4
Amazon Macie	ML-powered PII scanning across all S3 buckets; findings sent to CloudWatch	—	R4

The open-source core of MWAA (Apache Airflow) and Glue (Apache Spark) directly addresses the concern around vendor lock-in: these tools run identically on-premises or on any other cloud provider, ensuring Manga retains full portability.

3.4 Consumption Layer

Service	Consumer	Replaces	Key Capability
Amazon QuickSight	Business users, management	Daily Excel reports	Self-service dashboards, SPICE in-memory engine, pay-per-session pricing
Amazon SageMaker	Data science team	No ML capability today	End-to-end ML: feature store, training, deployment, monitoring
Amazon Comprehend	Merchandising, product teams	Manual review reading	Managed NLP: sentiment analysis on customer reviews at scale
API Gateway + Lambda	Partners, AI agents, 3rd parties	No external data access	Versioned REST APIs, rate limiting, usage plans

3.5 Cross-Cutting Services

Service	Purpose	RFP Req.
IAM + AWS KMS	Least-privilege access control; encryption at rest (S3, Redshift, DynamoDB) and in transit (TLS)	R4
Terraform / AWS CDK	IaC for all infrastructure; single source of truth provisioning Dev / Pre-Prod / Prod from identical templates	R2, R3
Amazon CloudWatch + SNS	Unified monitoring, custom metrics, pipeline health dashboards, PagerDuty alerting for on-call team	R4
AWS Organizations	Three separate accounts under one organisation; SCPs prevent accidental prod changes; per-account cost limits	R2
AWS Carbon Footprint Tool	Tracks emissions per service and region; year-over-year dashboard; quantifies improvement vs colocation baseline	R8

4. PROJECT IMPLEMENTATION

This section describes how the Manga DataHub proposal meets each stated RFP requirement, and presents the phased implementation methodology, timeline, and risk management approach.

4.1 Requirements Traceability

Req.	Requirement	How Addressed
R1	Unified multi-modal platform	S3 Lakehouse + Redshift + DynamoDB store structured, semi-structured, and unstructured data. Kinesis handles streaming; Glue handles batch. Single Glue Catalog provides unified access.
R2	Multi-environment setup	AWS Organizations provisions Dev, Pre-Prod, Prod as isolated accounts from identical Terraform templates. Glue PII masking + Lake Formation replicate anonymised samples to non-prod automatically.
R3	Automation	MWAA (Airflow) replaces all cron jobs with monitored DAGs. Terraform/CDK automates all infra. Glue job bookmarks, S3 lifecycle policies, and auto-scaling automate the data lifecycle end-to-end.
R4	Security & high availability	IAM least-privilege, KMS encryption, Lake Formation RBAC, Macie PII scanning, multi-AZ deployments for all critical services, CloudWatch alerting. Target: 99.9% uptime (AWS SLA backed).
R5	Data quality	Glue Data Quality rules run post-pipeline: null constraints, range validation, referential integrity, freshness thresholds. Failures trigger CloudWatch + SNS alerts before bad data reaches downstream.
R6	Cost optimisation	Pay-per-use vs fixed colocation. S3 Intelligent-Tiering, Spot Instances for Glue, Redshift Reserved Instances, QuickSight pay-per-session. Projected 40–60% infrastructure cost reduction in Year 1.
R7	Extensibility & interoperability	Decoupled Lambda connectors add sources without pipeline changes. API Gateway exposes curated data externally. Open Parquet/Airflow formats prevent lock-in. Versioned APIs ensure backward compatibility.
R8	Sustainability	EU (Ireland) / EU (Frankfurt) regions selected for renewable energy mix. Serverless (Lambda, Firehose) and auto-scaling eliminate idle resources. AWS Carbon Footprint Tool provides quantitative emission metrics vs colocation.

4.2 Implementation Methodology

CloudCore Solutions applies a **DataOps methodology**: iterative, test-driven, and with continuous integration for both data pipelines and infrastructure code. All pipeline code is version-controlled in Git. Data contracts are defined and validated between producer and consumer layers. Infrastructure changes undergo peer review before deployment. A dedicated data quality gate prevents promotion from Curated to Serving zones until quality thresholds are met.

The project is structured in three phases across nine months, with production cutover in Month 7 and full handover by Month 9.

4.3 Implementation Timeline

Phase	Duration	Key Deliverables	Milestone
Phase 1: Foundation	Months 1–3	AWS account structure (Organisations), Terraform IaC baseline, Dev environment live, DMS historical data migration, Glue Catalog populated, MWAA Airflow deployed, first batch pipelines operational	Dev environment fully operational
Phase 2: Build & Validate	Months 4–6	Pre-Prod environment live, Kinesis streaming pipelines active, Glue Data Quality rules deployed, Lake Formation RBAC configured, QuickSight dashboards replacing Excel, first ML model (demand forecasting) trained	Streaming + BI live in Pre-Prod; Excel decommissioned
Phase 3: Prod Cutover & ML	Months 7–9	Production environment live, full data cutover from colocation, recommendation engine deployed, dynamic pricing model live, security audit completed, team training delivered, full operations handover	Production live; colocation contract terminated

4.4 Risk Register

Risk	Probability	Impact	Mitigation
Data migration data loss	Low	High	DMS CDC replication runs parallel for 4 weeks before cutover; automated reconciliation checks compare row counts and checksums.
Vendor lock-in concerns (CTO)	Medium	Medium	Open-source tools at every layer (Airflow, Spark, Parquet). Architecture documented to be cloud-portable. CTO reviews all tool selections before sign-off.
GDPR non-compliance	Low	Critical	Lake Formation + Macie deployed in Phase 1, not Phase 3. GDPR DPIA completed before any personal data touches the new platform.
AWS cost overrun in Dev/Test	Medium	Low	Per-account cost budgets in AWS Organisations with automatic alerts at 80% threshold. Spot Instances and auto-off schedules for non-prod workloads.

5. SAMPLE USE CASES

The following three use cases demonstrate the Manga DataHub architecture in action, using Manga's actual data sources. Each illustrates a complete data engineering lifecycle from ingestion through to business outcome.

5.1 Demand Forecasting

Problem: Manga's buying and replenishment team currently relies on intuition and manual Excel analysis to estimate future demand. Stockouts cost revenue; overstock drives markdowns. The lack of external context (weather, local events, campaigns) makes seasonal planning particularly unreliable.

Data Sources Used:

- sales_sample.csv — historical transaction data providing the time-series signal
- external_factors_sample.csv — weather, bank holidays, local events (e.g. Fashion Week, Concerts), fuel prices, campaign IDs
- inventory_sample.csv — current stock levels per product per store

Pipeline Flow:

1. Glue ETL joins sales history with external_factors data, creating enriched time-series feature sets partitioned by store and product.
2. Features written to the SageMaker Feature Store in the Serving Zone.
3. SageMaker DeepAR (probabilistic time-series algorithm) trains weekly via an MWAA DAG. Model output: 7, 14, and 28-day demand forecasts per SKU per store.
4. Forecast results written to Redshift Serving Zone and surfaced in a QuickSight replenishment dashboard.
5. Automated replenishment orders triggered when forecasts breach minimum stock thresholds, integrating with the ERP via API Gateway.

Business Impact: **Projected 30% reduction in stockouts and 20% reduction in overstock write-offs** based on retail industry benchmarks for ML-driven replenishment.

5.2 Product Recommendation Engine

Problem: Manga's e-commerce site currently shows the same product listing to all visitors. Personalised recommendations — a standard feature on every competitor's platform — are unavailable because the data infrastructure cannot support real-time ML inference. Online sales are growing at only 1.2% YoY.

Data Sources Used:

- sales_sample.csv — purchase history per customer
- customers_sample.csv — customer demographics and loyalty programme status
- inventory_sample.csv — current stock levels (ensures recommendations are in-stock)

Pipeline Flow:

6. Real-time purchase events stream through Kinesis and update a customer-product interaction matrix in the SageMaker Feature Store.
7. SageMaker trains a collaborative filtering model weekly on the full interaction history. Model outputs: top-10 product recommendations per active customer.
8. Recommendations are filtered against live inventory data (DynamoDB) to ensure only in-stock items are surfaced.

9. Results cached in DynamoDB. The e-commerce platform calls the API Gateway endpoint ($\leq 5\text{ms}$ response) to retrieve personalised recommendations per session.

Business Impact: Retail industry benchmarks for personalised recommendations show **12–18% increase in average order value (AOV)** and **25–35% improvement in click-through rates** on recommended products.

5.3 Sentiment-Driven Merchandising Dashboard

Problem: Manga's buying team has no systematic way to monitor customer sentiment on products. Negative trends go undetected until they appear in sales declines, by which time inventory has already been committed.

Data Sources Used:

- `customer_reviews_sample.csv` — review text, star ratings, product and customer IDs, timestamps, image uploads

Pipeline Flow:

10. Review submission events stream via Kinesis in real-time as customers complete reviews on the website or app.
11. Amazon Comprehend processes review text, returning sentiment scores (Positive / Negative / Neutral / Mixed) and key phrase extraction per review.
12. Comprehend output is joined with product and store metadata in the Curated Zone. Glue aggregates sentiment scores by product, store, and rolling 7-day window.
13. QuickSight dashboard surfaces products with a week-over-week sentiment decline ≥ 0.15 points. SNS alert triggers automatically when a product crosses the negative threshold.
14. Buying team receives actionable alerts 2–3 weeks before negative sentiment impacts sales figures.

Business Impact: **Average 2–3 week early warning** on product quality issues. Reduces reactive markdown spend and strengthens customer trust through faster product improvement cycles.

6. FUTURE IMPROVEMENTS

The Manga DataHub is designed to be extensible. The following roadmap presents enhancements beyond the initial 9-month delivery, ordered by implementation horizon and strategic impact.

6.1 Short-Term (Months 10–18)

Real-Time Personalisation API

Extend the recommendation engine to serve real-time, session-level personalisation across the website and in-store digital touchpoints. Leverages the existing Kinesis + DynamoDB infrastructure. Enables Manga to deliver Amazon-style “Customers who bought this also bought” experiences at sub-10ms latency.

Supply Chain Smart Labels Integration

Integrate RFID and smart label data from logistics partners into the Kinesis streaming pipeline. Enables real-time inventory tracking at the item level across the entire supply chain, reducing shrinkage and enabling same-day stock replenishment. Directly addresses the smart labels use case listed in the RFP.

Advanced Data Quality — Expectation Suites

Extend Glue Data Quality with Great Expectations-style expectation suites per dataset, including statistical distribution checks that flag when daily sales volume deviates more than 3 standard deviations from the 30-day rolling average.

6.2 Medium-Term (Months 18–30)

Virtual Shopping Assistant — Amazon Bedrock

Deploy a retrieval-augmented generation (RAG) powered shopping assistant using Amazon Bedrock (Claude or Titan foundation models) grounded on Manga’s product catalog, inventory data, and customer purchase history. Directly addresses the “Virtual shopping assistant” use case listed in the RFP.

External Data Monetisation via API Marketplace

Expose anonymised, aggregated retail trend data (sales velocity by category, regional demand signals) to external consumers via a managed API marketplace. Transforms the data platform from a cost centre into a revenue-generating asset.

Data Mesh Architecture

As Manga’s data team grows across business units, transition from a centralised data platform to a data mesh model: individual business domains (merchandising, logistics, marketing) own and publish their data products through the shared Glue Catalog and Lake Formation governance layer.

6.3 Long-Term (30+ Months)

AI Agent Ecosystem

The RFP explicitly identifies “data exposure to external consumers, including AI agents” as a target use case. Amazon Bedrock Agents provides the managed infrastructure for building and deploying Manga’s own internal agents that act across the platform autonomously, and for exposing structured data endpoints to external agent ecosystems.

Federated ML Across Retail Partners

Implement federated learning models that train across data from Manga's network of logistics and supply chain partners without sharing raw data. Enables collaborative demand forecasting with supplier visibility while maintaining full data sovereignty.

Full Carbon Neutrality Roadmap

Build on the AWS Carbon Footprint Tool metrics from Day 1 to establish a credible, board-reportable pathway to carbon neutrality. Includes workload scheduling to run compute-intensive jobs during renewable energy surplus windows, and AWS Graviton processor migration for energy-efficient compute.

APPENDIX

A. References

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B. Generative AI Acknowledgement

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C. Team Contributions

Team Member	Primary Responsibilities
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Tina Jannasch	Low-Level Architecture (Section 3), AWS service selection and cost rationale
Ricardo Liévano Pedroza	Project Implementation (Section 4), R1–R8 requirements mapping, risk register, timeline
Luka Tcheishvili	Data sources analysis, Sample Use Cases (Section 5), pipeline design, interactive Streamlit app
Nicklas Urban	Future Improvements (Section 6), GitHub repository, deployment, presentation lead